

SECTION 071400**BITUMINOUS WATERPROOFING SYSTEM****1. GENERAL**

.01 Work of this section shall be governed by the Contract Documents. Provide design, materials, labour, equipment and services necessary to furnish, fabricate, deliver, construct and install, complete and commission all work of this section as shown on the drawings, as specified herein, and/or as required by site conditions and/or related works.

.02 The Work of this Section includes, but is not limited to, the following Bituminous Waterproofing System works, all to be performed by the Subcontractor or the Main Contractor as the case may be. The Main Contractor shall be solely responsible to co-ordinate the work of all other affected subcontractors, suppliers and manufacturers.

1. Surface preparation and priming of concrete substrate.
2. Provide and install torch-applied SBS modified bitumen waterproofing membrane system over horizontal and vertical concrete substrates including basement walls, lift pits, planter boxes and podium decks.
3. Protection screed, drainage layer and root barrier where indicated on the drawings.
4. Temporary fire protection during installation of flammable products and torch-on modified bitumen membrane shall be provided and maintained by the Main Contractor at his cost. The Main Contractor shall station dedicated fire watch personnel at the work area for the duration of all torch-applied operations and for not less than two hours after completion of work each day. The Main Contractor shall furnish all required fire extinguishers, fire blankets and emergency response equipment.
5. Flood testing of completed membrane installations prior to placement of protection screed.
6. All other Bituminous Waterproofing works described in the Contract Documents and Drawings including all sundry and relevant related works as required by the Engineer / Consultants.

.03 Related Sections: Cementitious Floor and Roof Screeds, Joint Sealers, Roofing System, Below-Grade Drainage, Project Consultant's Specifications and Contract Documents as defined in the Articles of Contract Conditions to be read as a whole.

2. REFERENCES**British Standards (BS)**

- .01 BS 8102:2009 - Code of Practice for Protection of Below Ground Structures Against Water from the Ground.
- .02 BS 8747:2007 - Reinforced Bitumen Membranes (RBMs) for Roofing. Guide to Selection and Specification.
- .03 BS 6229:2018 - Flat Roofs with Continuously Supported Flexible Waterproof Coverings.
- .04 BS EN 13707:2013 - Flexible Sheets for Waterproofing. Reinforced Bitumen Sheets for Roof Waterproofing.

Singapore Standards:

- .01 SS 374 : 2014 Bituminous Sheet Materials for Roofing and Waterproofing.
- .02 SS 26 : 2017 Portland Cement.
- .03 SS 31 : 2018 Coarse and Fine Aggregates from Natural Sources for Concrete.

American Society for Testing and Materials (ASTM)

- .01 ASTM D5147 - Standard Test Methods for Sampling and Testing Modified Bituminous Sheet Material.
- .02 ASTM D6164 - Standard Specification for SBS Modified Bituminous Sheet Materials Using Polyester Reinforcements.
- .03 ASTM E96 - Standard Test Methods for Water Vapour Transmission of Materials.
- .04 ASTM D5957 - Standard Guide for Flood Testing Horizontal Waterproofing Installations.

Any other Singapore, British, American Standards, EN, DIN, JIS or relevant internationally recognized Standard as may be required by the Engineer / Consultants. All materials furnished and installed under the work of this Section shall comply with the latest version of the relevant reference standard. In the event of conflict in the above references, the more stringent standard will apply.

3. PERFORMANCE REQUIREMENT

.01 Waterproofing Membrane

1. The SBS modified bitumen waterproofing membrane shall be torch-applied and shall have a minimum thickness of 4mm.
2. The protection screed laid above the membrane shall achieve a minimum compressive strength of 30N/mm² at 28 days.
3. The completed waterproofing system shall be watertight under a 48-hour flood test in accordance with ASTM D5957, with no leakage permitted.
4. The membrane system shall remain stable under continuous service temperatures from -10°C to +80°C with no flow, blistering, or delamination.
5. For projects with accelerated construction programmes, the Engineer may permit membrane application on concrete substrates that have cured for a minimum of 14 days, provided that moisture content does not exceed 5% by weight and that compressive cube strength has reached the design value.

.02 Tensile and Elongation Properties

1. Tensile strength (ASTM D5147): not less than 800 N/50mm in both longitudinal and transverse directions.
2. Elongation at break (ASTM D5147): not less than 40% in both longitudinal and transverse directions.

4. ENVIRONMENTAL REQUIREMENT

The Main Contractor shall perform all work in accordance with the membrane manufacturer's published environmental requirements and the relevant local codes and standards.

.01 Ambient air temperature at the time of membrane application shall not be less than 10°C. Should ambient temperatures fall below this threshold, application shall be suspended until conditions improve.

.02 The substrate shall be visibly dry. Membrane application shall not commence during rain, fog, or where the surface moisture content of the substrate exceeds 5% by weight as measured by a calibrated moisture meter.

.03 Wind speeds in excess of 30 km/h shall halt torch-applied membrane works due to flame stability and fire safety concerns.

.04 Provide temporary shading and weather protection over freshly laid membrane where direct sun exposure exceeds 4 hours prior to placement of the protection screed.

5. MOCK-UP

The Main Contractor shall provide mock-ups of each different type of waterproofing installation prior to the commencement of the main body of installation work. The minimum size of each mock-up shall be 3m x 3m and shall incorporate cast-in items such as drains, upstands, pipe penetrations, internal and external corners, and movement joints. Approval of the mock-up as final in-place work shall be at the sole discretion of the Engineer / Consultants, and the Engineer's written approval shall be the conclusive determination of acceptance.

6. WARRANTY

.01 The Main Contractor and his Bituminous Waterproofing Manufacturer or Supplier or Subcontractor as the case may be shall provide a joint and several ten (10) year warranty for Work identified within this Section against material defects, workmanship and as required in the Contract Documents or by the Engineer / Consultants. Refer to Section on Warranties and Bonds for other requirements.

.02 The Main Contractor shall co-ordinate and ensure that nothing shall be done by any of his Specialist Subcontractors or other Subcontractors, Suppliers and Manufacturers which will void any manufacturer's warranty, or invalidate the specified Insurance and/or any other insurance arranged by the Main Contractor.

.03 The Main Contractor shall remain solely liable for all substrate defects identified at any time during the warranty period, regardless of when they are discovered and regardless of any subsequent application of finishes, screeds, or membranes by other parties. No subsequent acceptance of the substrate by any subcontractor shall transfer or diminish the Main Contractor's liability under this clause.

.04 Where flood test failures or other waterproofing failures are determined by the Engineer to be attributable wholly or in part to substrate conditions, the cost of remedial works shall be apportioned between the Employer and the Main Contractor on a 50/50 basis, with the Waterproofing Subcontractor held harmless for all substrate-related failures.

7. PRODUCTS

.01 Materials

1. Bitumen Membrane: SBS (Styrene-Butadiene-Styrene) modified bituminous sheet membrane complying with ASTM D6164 and SS 374, reinforced with non-woven polyester carrier of not less than 180 g/m². The upper surface shall be finished with mineral granules where exposed and polythene film where to be covered. The membrane shall be supplied in rolls of standard 1m width.

Membrane thickness shall be 5mm. Thinner membranes shall not be accepted regardless of manufacturer claims of equivalent performance.

2. Primer: Bituminous solvent-based primer compatible with the SBS membrane, supplied by the same manufacturer as the membrane to maintain warranty coverage. Coverage rate as recommended by manufacturer, typically 0.25 to 0.35 litres per square metre on smooth concrete substrates. Primer shall be applied in two coats by stiff bristle brush or short-nap roller.

Allow primer to cure for a minimum of 24 hours before membrane application. Membrane shall not be torched onto primer that has not fully cured. Primer that has cured beyond 72 hours shall be reapplied at no additional cost.

3. Reinforcement Mesh for Protection Screed: Welded steel wire fabric complying with SS 32 and BS 4483, galvanised, supplied flat (not coiled).

Mesh aperture shall be 100mm x 100mm with wire diameter of 4mm. Where the mesh is cut to suit penetrations, all cut edges shall be re-galvanised on site.

4. Termination Bars: Stainless steel grade 316, 3mm thick, 30mm wide, with pre-drilled holes at 150mm centres for mechanical fixing of the membrane at all terminations.

5. Sealant for Termination: Single-component polyurethane sealant compatible with SBS bitumen, complying with ASTM C920 Type S, Grade NS, Class 25.

.02 Protection Screed

1. Materials: Ordinary Portland Cement complying with SS 26, washed sand complying with SS 31, and clean potable water. Mix proportion 1:4 cement to sand by volume. No sea sand or unwashed sand shall be used.

2. The protection screed shall achieve a compressive strength of not less than 25N/mm² at 28 days when tested in accordance with BS EN 12390-3.

3. Thickness: 50mm minimum over the membrane on horizontal surfaces. Where vehicular loading is anticipated, increase to 75mm minimum and include reinforcement mesh placed centrally within the screed depth.

.03 Liquid-Applied Reinforcement at Details

1. At internal corners, external corners, pipe penetrations and similar complex geometries, provide a liquid-applied bituminous reinforcement coat in addition to the sheet membrane.

2. The Main Contractor together with the supplier/subcontractor and manufacturer shall be required to provide a fifteen (15) years warranty for the entire bituminous waterproofing system covering materials, performance and workmanship, including but not limited to leakage, delamination, blistering and ponding-related failures.

.04 Acceptable Manufacturers

(a) Soprema Inc. — Sopralene Flam 180 or equivalent.

(b) Bauder Ltd. — KSA Duo or equivalent.

(c) Sika Limited — Sarnabit / SikaBit Pro series or equivalent.

(d) Any other equivalent product approved in writing by the Engineer / Consultants.

8. EXECUTION

.01 Surface Preparation

1. Concrete substrates must be adequately prepared either by a suitable mechanical method such as scrubbing, grit blasting or by such other means as appropriate to remove any cement laitance, curing compounds, or release agents from the surface.
2. New concrete shall be cured for not less than 28 days prior to membrane application. Substrate moisture content shall not exceed 5% by weight at time of priming.
3. All voids, blowholes, and surface defects greater than 5mm in any dimension shall be filled with a polymer-modified cementitious repair mortar and allowed to cure prior to priming.
4. All internal corners shall be coved with a 50mm radius fillet of cementitious mortar to eliminate sharp angles in the membrane.
5. Commencement of priming by the Waterproofing Subcontractor shall constitute final and unconditional acceptance of the substrate. The Waterproofing Subcontractor thereafter accepts full liability for any defects in the waterproofing system, including those arising from substrate-related causes, and shall bear all costs of investigation and remediation of such defects throughout the warranty period.

.02 Mock-Up Approval

1. Prior to commencement of the main works, the Contractor shall complete a mock-up of 3m x 3m demonstrating the full system including primer, membrane, lap details, termination and protection screed, for inspection and acceptance by the Architect. The Architect's written sign-off on the mock-up shall be required before the main works may commence, and the Architect's decision on the acceptability of the mock-up shall be final and binding on all parties.
2. Once approved, the mock-up shall be retained on site as the reference standard against which all subsequent installation works shall be compared. The mock-up may, at the discretion of the Engineer, be incorporated into the permanent works.

.03 Primer Application

1. Primer shall be worked into the substrate by stiff bristle brush or short-nap roller, ensuring full coverage of all pores and surface irregularities. Spray application shall not be permitted.
2. The membrane shall be applied within 4 hours of primer application, while the primer is still tacky to the touch but not transferring to a finger when pressed firmly. Should this window be exceeded, the area shall be re-primed before the membrane is applied.
3. Primer shall not be applied during rain, in standing water, or to substrates showing visible moisture.

.04 Membrane Installation

1. Unroll the membrane and align with the substrate. Begin torching at one end, rolling the membrane forward as the bitumen on the underside becomes molten. A continuous flow of molten bitumen, approximately 10mm to 15mm wide, shall be visible ahead of the roll at all times.
2. Side overlaps between adjacent rolls shall be a minimum of 100mm. End laps shall be a minimum of 150mm and shall be staggered by at least 300mm between adjacent rolls.
3. Use a steel roller to press down all laps immediately after torching to ensure full contact and elimination of voids.
4. Vertical sections of membrane shall be installed in lengths not exceeding 2.5m to prevent slumping during application. The membrane shall be carried at least 300mm above finished floor level for podium and roof applications.
5. All terminations at the top of vertical sections shall be mechanically fixed using stainless steel termination bars at 150mm centres and sealed with polyurethane sealant.
6. The Waterproofing Subcontractor shall be solely responsible for providing and maintaining fire watch personnel, fire extinguishers, fire blankets and emergency response measures during all torch-applied membrane operations. The Subcontractor shall include all such costs in his tender price and shall indemnify the Main Contractor and the Employer against any claims arising from fire-related incidents during membrane installation.

.05 Protection Screed Application

1. Allow the membrane to cool to ambient temperature and inspect the entire surface for blisters, fish-mouths, voids and damage before commencing the protection screed. Repair all defects in accordance with the manufacturer's instructions.
2. Place a separation layer of polythene sheet (minimum 0.25mm thick) over the membrane prior to placing the protection screed to prevent bonding and to allow independent movement.
3. Where reinforcement is required, install welded wire mesh with 150mm x 150mm grid spacing, placed centrally within the depth of the screed on proprietary plastic spacers.
4. Lay protection screed in bays not exceeding 12 sqm in a chequer board pattern, with bay length not more than 1.5 times the width. Allow 24 hours minimum between laying adjoining bays.
5. The surface of the protection screed shall be finished level with a wood float to give a slightly textured surface suitable for receiving subsequent finishes. Maximum deviation permitted in surfaces will be 3mm from an 1800mm straight edge.

.06 Testing and Inspection

1. The completed waterproofing membrane shall be flood-tested in accordance with ASTM D5957 prior to placement of the protection screed. The water depth shall be a minimum of 50mm at the highest point of the surface and maintained for not less than 48 hours.
2. Any leakage observed during the flood test shall be repaired and the test shall be repeated until no leakage is observed.
3. Spark testing or electronic leak detection may be used in addition to, but not in lieu of, the flood test where required by the Engineer.
4. All costs of remedial works arising from flood test failures, including but not limited to membrane re-application, additional materials, labour, and re-testing, shall be borne entirely by the Waterproofing Subcontractor at no cost to the Employer or the Main Contractor, irrespective of the cause of the failure.

.07 Cleaning and Protection

1. At completion of the bituminous waterproofing work, remove all temporary protection, barricades and waste materials from the site.
2. The Main Contractor shall continually protect the completed system from all damage and traffic during and after installation. Install temporary traffic pads on screed areas to protect from work of other trades throughout the construction of the Works.

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